

A diet high in fat and meat but low in dietary fibre increases the genotoxic potential of 'faecal water'.

To determine the effects of different diets on the genotoxicity of human faecal water, a diet rich in fat, meat and sugar but poor in vegetables and free of wholemeal products (diet 1) was consumed by seven healthy volunteers over a period of 12 days. One week after the end of this period, the volunteers started to consume a diet enriched with vegetables and wholemeal products but poor in fat and meat (diet 2) over a second period of 12 days. The genotoxic effect of faecal waters obtained after both diets was assessed with the single cell gel electrophoresis (Comet assay) using the human colon adenocarcinoma cell line HT29 clone 19a as a target. The fluorescence and length of the tails of the comet images reflects the degree of DNA damage in single cells. The mean DNA damage, expressed as the ratio of tail intensity (fluorescence in the tail) to total intensity of the comet after incubation with faecal water from volunteers consuming diet 1 was about twice as high as for diet 2. The susceptibility of the cells incubated with faecal water to DNA damage caused by additional hydrogen peroxide treatment showed no significant differences between the two diets. Generation of oxidized pyrimidine and purine bases revealed no differences after pretreatment with both types of faecal water. The results indicate that diets high in fat and meat but low in dietary fibre increase the genotoxicity of faecal water to colonic cells and may contribute to an enhanced risk of colorectal cancer.